Vector Databases: Indexing

A flat index stores every vector and scans all of them at query time, which is exact but slow as the collection grows. FAISS provides this as IndexFlatIP, using inner product over normalized vectors so the score equals cosine similarity. It is the right default for a few thousand chunks where exact search is fast enough and simplicity matters.

Vector Databases: Approximate Search

For millions of vectors an exact scan is too slow, so approximate nearest neighbour indexes such as HNSW and IVF trade a small amount of recall for a large speedup. HNSW builds a navigable small-world graph and reaches the neighbourhood of a query in logarithmic time. Choosing the index is a recall-versus-latency trade-off, not a correctness one.